For a *fixed amount* of gas

$$\boxed{\frac{pV}{T} = \text{constant}} \quad \text{i.e.,} \quad \frac{p_1V_1}{T_1} = \frac{p_2V_2}{T_2} \qquad p = \frac{nRT}{V} \propto \frac{1}{V}$$

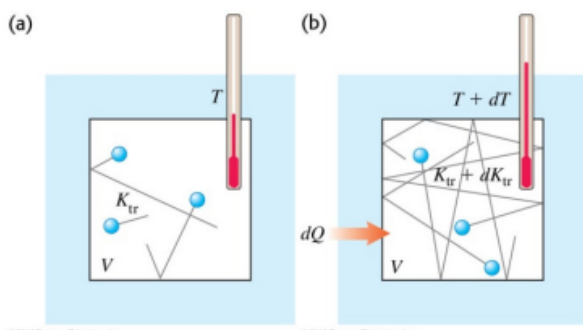
called **Boyle's law**

ideal gas law: $\boxed{pV = nRT = NkT}$

total translational KE of ideal gas: $\boxed{K_{\text{tr}} = \frac{3}{2}nRT = \frac{3}{2}NkT}$

Heat Capacity of Ideal Gas

At constant volume, i.e., no expansion, C_V **molar heat capacity at constant volume**



heat needed to raise temperature by dT

$$dQ = nC_V dT$$

increase in KE of gas

$$dK_{\text{tr}} = \tfrac{3}{2}nRdT$$

$$nC_V dT = \tfrac{3}{2}nRdT$$

$$\Rightarrow \quad \boxed{C_V = \frac{3}{2}R} = 12.47 \text{ J/mol·K}$$

For an ideal gas, $U = \frac{f}{2}NkT \propto T$

First Law of Thermodynamics

$$\boxed{Q = \Delta U + W}$$

Suppose an ideal gas changes volume at *constant pressure*

$$W = \int_{V_1}^{V_2} pdV = p(V_2 - V_1)$$

at *constant temperature*

$$W = \int_{V_1}^{V_2} pdV = \int_{V_1}^{V_2} \frac{nRT}{V} dV = nRT \ln\frac{V_2}{V_1} = nRT \ln\frac{p_1}{p_2}$$

Isochoric process means constant volume, e.g., in a close constant-volume container
Importance: pdV work is zero

Isobaric process means constant pressure, e.g., in atmospheric pressure
Importance: pdV work is $p(V_2 - V_1)$

Isothermal process means constant temperature, i.e., *isotherms* in a pV diagram

Adiabatic process means $Q = 0$, it may be

1. a thermally insulated system where heat cannot enter or leave, e.g. in a thermal flask
2. a quick process where heat has no time to go in or out, e.g., by conduction

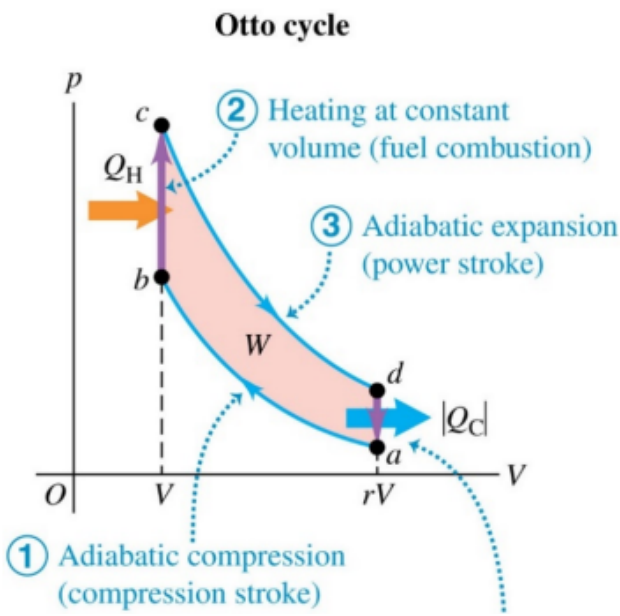
Adiabatic *expansion*, $W > 0 \Rightarrow \Delta U < 0$, usually leads to *cooling*

Same temperature → $\Delta U = 0$

$$\boxed{C_p = C_V + R} \qquad C_V = \frac{R}{\gamma - 1}$$

Define the **thermal efficiency** of the engine

$$\boxed{e = \frac{W}{|Q_\mathrm{H}|} = 1 - \left|\frac{Q_\mathrm{C}}{Q_\mathrm{H}}\right|}$$



For the isochoric processes *bc* and *da*

$$Q_H = nC_V(T_c - T_b) > 0$$
$$Q_C = nC_V(T_a - T_d) < 0$$

therefore

$$e = \frac{Q_H + Q_C}{Q_H} = 1 - \frac{T_d - T_a}{T_c - T_b}$$

For the adiabatic processes *ab* and *cd*

$$T_a(rV)^{\gamma-1} = T_b V^{\gamma-1}$$
$$T_d(rV)^{\gamma-1} = T_c V^{\gamma-1}$$

Hence

$$\boxed{e = 1 - \frac{1}{r^{\gamma-1}}} < 1$$

Define **coefficient of performance**

$$\boxed{K = \frac{|Q_C|}{|W|} = \frac{|Q_C|}{|Q_H| - |Q_C|}}$$

$$\boxed{e_{\text{Carnot}} = 1 - \frac{T_C}{T_H} = \frac{T_H - T_C}{T_H}}$$